"On my honor I have neither given nor received any unauthorized and or inappriprate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

Mehmet KAYA
150 719 564

1-) $A = \begin{bmatrix} 1 & 2 & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & -3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \xrightarrow{-R_1+R_3} \begin{bmatrix} 1 & 2 & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & -3 & \alpha^2 \\ 0 & -\alpha+14 & \gamma-6 & -\alpha+2 \end{bmatrix}$

$\xrightarrow{-2R_1+R_3} \begin{bmatrix} 1 & 2 & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 0 & -2\alpha+\beta & +2\gamma-3 & -2\alpha+\alpha^2 \\ 0 & -\alpha+14 & \gamma-6 & -\alpha+2 \end{bmatrix}$

$1 + \alpha - \gamma = \alpha \Rightarrow \gamma = 1 //$

$3\alpha + 5 - 9\gamma = 1$

$3\alpha = 5 \quad \alpha = 5/3 //$

$-2\alpha + \beta + 2\gamma - 3 = 2\alpha + \alpha^2$

$-\frac{10}{3} + \beta + 2 - 3 = \frac{30}{9} + \frac{25}{9}$

$-\frac{29}{9} + \beta = \frac{55}{9} \qquad \beta = \frac{94}{9} //$

$k = -2 // \quad m = 0 //$

② a-) Since the vector space with vector satisfying $x = 5y$ which means $x - 5y \Rightarrow$ is of dimension $R^n$ must be of dimension $n \geq 5$ such that $x \leq R^5$

$x - 5y \Rightarrow \quad K \subseteq R^5$

3-) Suppose that u is a unit vector in $\mathbb{R}^n$ ...

$H = I - 2uu^T$

(a-) $H^2 = H \cdot H = (I - 2uu^T) \cdot (I - 2uu^T)$

$= I^2 - (I \cdot 2uu^T) - (2uu^T \cdot I) + (4uu^T \cdot uu^T)$

$= I(I - 2uu^T - 2uu^T) + 4u\,\|u\|^2 \cdot u^T$

$= I - 2uu^T - 2uu^T - 4u \cdot \|u\| \cdot u^T$

$\|u\|^2 = u^T \cdot u$

$\|u\| = \sqrt{u^T \cdot u}$

$\Rightarrow \|u\| = 1$

So, $H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4(1) + 4(1)$

$\boxed{H^2 = I_{//}}$

(b-) $HH^T = I - (2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T) \cdot (I^T - 2(u^T)^T u^T)$

$= (I - 2uu^T) \cdot (I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T \cdot uu^T$

$= I - 4uu^T + 4u\,\|u\|^2 u^T$

$= I - 4uu^T + 4uu^T$

$= I - 4(1) + 4(1)$

$= I_{//}$

$I = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}_{//}$

So $\Rightarrow HH^T = I$

$H^T H = I$

So, "H" is orthogonal matrix.

(c-) For a matrix A, a vector will be eigen vector of A corresponding eigen value $\lambda$ if $\underline{Av = \lambda v}$.

$$Hu = (I - 2uu^T) \cdot u$$
$$= Iu - 2uu^T u$$
$$= u - 2u \cdot \|u\|^2$$

I is Identity matrix. So, 1 is the only eigen value of I and,

$Iu = 1 \cdot u = u$ and, $\|u\| = +\sqrt{u^T \cdot u}$

u is a unit vector so, $\|u\| = 1 = +\sqrt{u^T \cdot u}$

Hence, $u^T u = \|u\|^2 = 1$

So, $uu = u - 2u = -u$

$\rightarrow Hu = -u = (-1)u$ //

$\Rightarrow$ Then we have $\rightarrow Hu = \lambda u$

$\lambda = -1$ //

So u is an eigenvector of H.

Corresponding eigen value for $u$ i.e. $\lambda = -1$

(d-) $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T = (I^T - (2uu^T)^T) = (I - 2(u^T)^T u^T)$

$= I - 2uu^T = H$ //

H is a symetric matrix, and available theorem which states that,

$H = Q \Lambda Q^T$

$= Q \Lambda Q^{-1}$

$= (Q^{-1})^{-1} \Lambda Q^{-1}$

$= P^{-1} \Lambda P$

Q is an orthogonal matrix,

$\Lambda$ = diagonal matrix

Thus, H is diagonalizable matrix.

(4-)

a-) $\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$ this diff. equation set expreces the process.